

结构实验模态理论

–实模态理论

清华大学航天航空学院

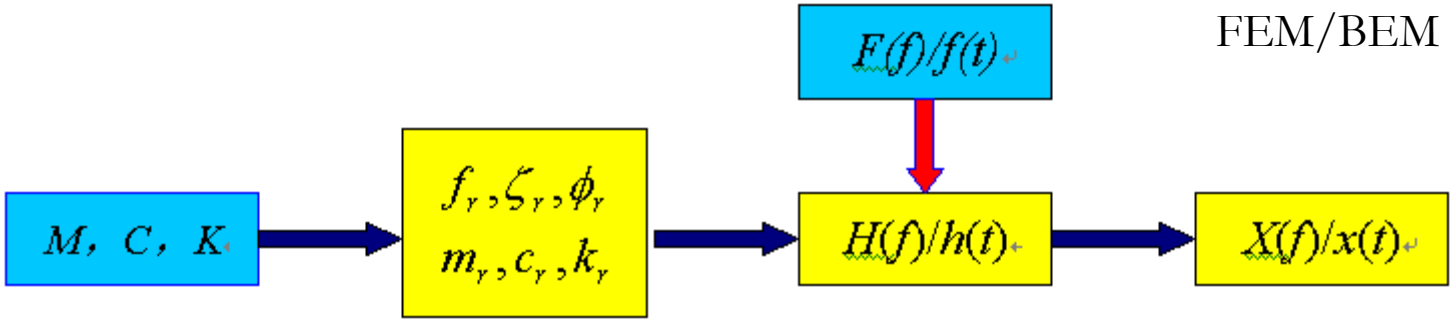
陆秋海

结构动力学特性分析的两种方法

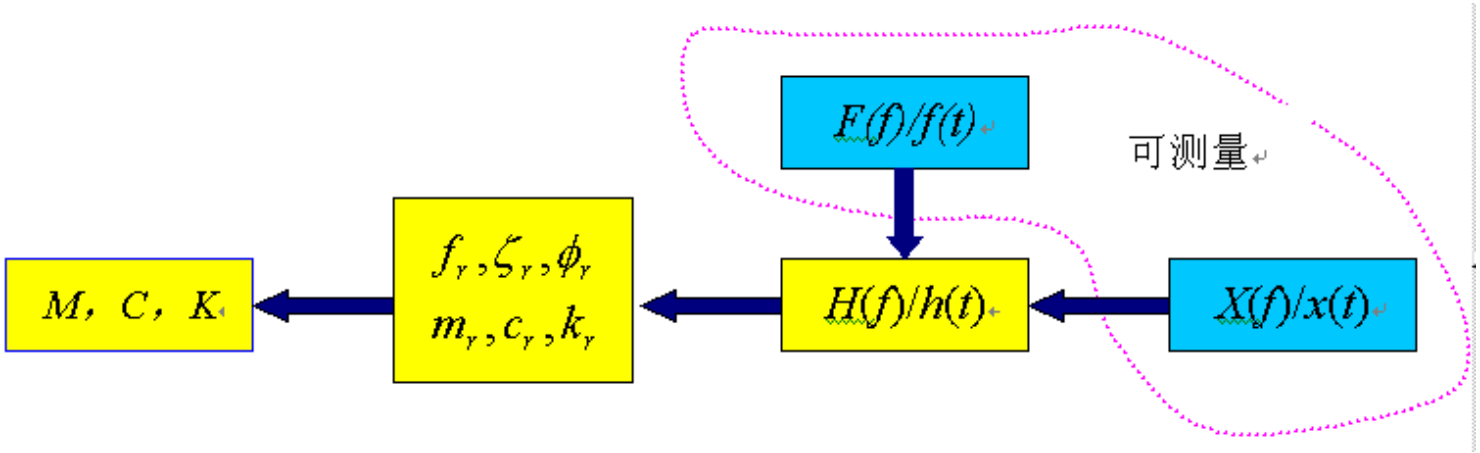
$$M\ddot{x} + C\dot{x} + Kx = f(t)$$

物理坐标　　模态坐标　　物理坐标

FEM/BEM



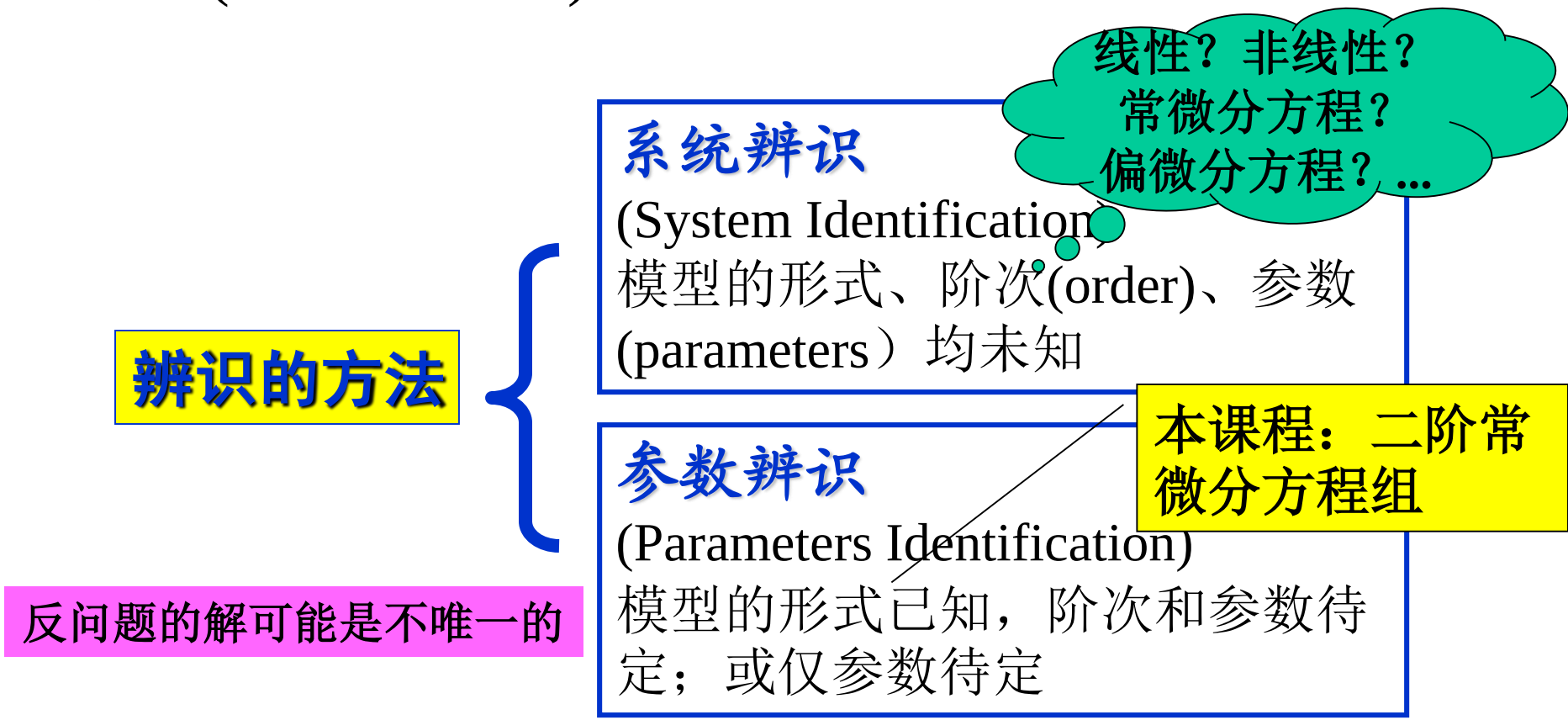
分析过程(Analytical Procedure)



辨识过程(Identification Proceduce)（模态未截断，激励已知）

1 实验建模及分类

实验建模 通过系统的**输入**(Input)和**输出**(Output),或仅通过**输出**信号,确定系统动特性模型的方法。又称为辨识(Identification)过程。



系统分类

离散多自由度线性系统运动方程

$$M\ddot{x}+C\dot{x}+Kx=f(t)$$

M,C,K分别为N×N维的系统离散质量矩阵、阻尼矩阵和刚度矩阵，x为N维广义位移矢量，$f(t)$为N维广义力矢量。

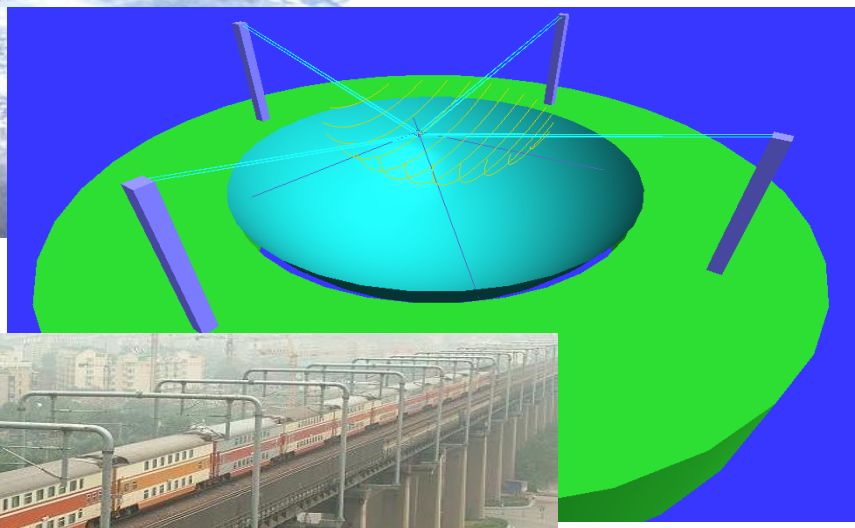
定常（时不变）系统
(Time Invariant System)
M,C,K矩阵均为与时间无关的常数矩阵

时变系统
(Time Varying System)
M,C,K不同时为与时间无关的常数矩阵。

时变系统(Time Varying System)举例









And what else?

2 实模态(Normal Mode Theory)

$$M\ddot{x}+C\dot{x}+Kx=f(t)$$

条件：

有限自由度N

阻尼可对角化

质量、刚度、阻尼阵实对称

质量阵可逆，即

$$M^{\mathrm{T}}=M,$$

$$C^{\mathrm{T}}=C,\quad M,C,K\in\mathbf{R}^{\mathrm{N}}\qquad \exists M^{-1}$$

$$K^{\mathrm{T}}=K,$$

2.1 特征值问题(Eigen Equation)

N维有阻尼线性系统振动方程(Motion Equation)

$$M\ddot{x}+\cancel{C\dot{x}}+Kx=\cancel{f(t)}$$

无阻尼自由振动方程(Free Vibration)

$$M\ddot{x}+Kx=0$$

令 $x=Xe^{j\omega t}$

广义特征值问题特征方程(General Form)

$$\omega^2 MX=KX$$

令 $A=M^{-1}K$

$\lambda=\omega^2$

标准特征值问题特征方程(Normal Form)

$$AX=\lambda X$$

特征对 $\omega_i \quad \phi_i, i=1,\ldots,N$

命题：实对称阵的特征值是实的。
推论：实对称阵的特征向量可以取实向量。

证明：

$A^{\mathrm{T}} = A;\quad A^{*} = A;\quad AX = \lambda X$

则

$$X^{\mathrm{H}}AX = (X^{*})^{\mathrm{T}}A^{\mathrm{T}}X = (AX^{*})^{\mathrm{T}}X = ((AX)^{*})^{\mathrm{T}}X = (AX)^{\mathrm{H}}X$$

所以

$$\lambda X^{\mathrm{H}}X = (\lambda X)^{\mathrm{H}}X = \lambda^{*}X^{\mathrm{H}}X$$

因为$X \neq 0$，所以

$$\lambda = \lambda^{*}$$

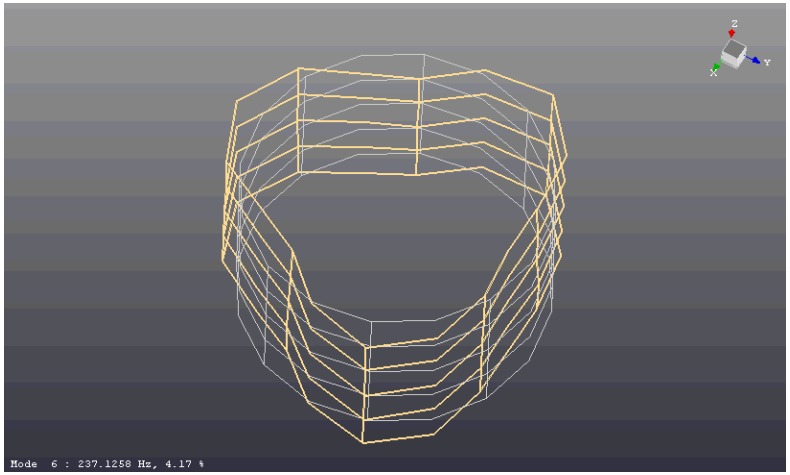
因为特征向量X是实系数齐次特征方程$(A - \lambda I)X = 0$的解，所以，则实对称阵的特征向量可以取实向量。

实模态假定：系统具有N对各异的实特征值和特征向量

特征值(Eigen Values)/模态频率(Modal Freqencies) ω_r r=1,…,N
特征向量(Eigen Vectors)/模态振型(Mode Shapes) ϕ_r

特征值和特征向量均为实数，即：

- 各点的**相位差相同**：同时过零点，同时达到最大值。
- 存在节点或节线



2.2 实模态的正交条件(Orthogonality of the Real Modes)

用于方程的解耦(decouple)，又称对角化条件

$$\{\varphi_k\}^{\mathrm{T}}\quad [K]\{\varphi_r\}=\Omega_r^2[M]\{\varphi_r\}$$

$$\{\varphi_k\}^{\mathrm{T}}[K]\{\varphi_r\}=\Omega_r^2\{\varphi_k\}^{\mathrm{T}}[M]\{\varphi_r\}$$

$$[K]\{\varphi_k\}=\Omega_k^2[M]\{\varphi_k\}\quad \mathbf{T}\{\varphi_r\}$$

$$-\{\varphi_k\}^{\mathrm{T}}[K]\{\varphi_r\}=\Omega_k^2\{\varphi_k\}^{\mathrm{T}}[M]\{\varphi_r\}$$

$$(\Omega_r^2-\Omega_k^2)\{\varphi_k\}^{\mathrm{T}}[M]\{\varphi_r\}=0$$

$r\neq k,\Omega_r\neq\Omega_k$

$$\{\varphi_k\}^{\mathrm{T}}[M]\{\varphi_r\}=0 \qquad \{\varphi_k\}^{\mathrm{T}}[K]\{\varphi_r\}=0$$

$r=k,\Omega_r=\Omega_k$

$$\{\varphi_r\}^{\mathrm{T}}[M]\{\varphi_r\}=m_r \qquad \{\varphi_r\}^{\mathrm{T}}[K]\{\varphi_r\}=k_r$$

广义质量，模态质量　　广义刚度，模态刚度

$$\{\varphi_r\}^{\mathrm{T}}[K]\{\varphi_r\}=\Omega_r^2\{\varphi_r\}^{\mathrm{T}}[M]\{\varphi_r\}$$

$$k_r=\Omega_r^2 m_r \qquad \Omega_r^2=k_r m_r^{-1}$$

固有振型的正交特性

$$[\Phi]^{\mathrm{T}}[M][\Phi]=[M_r],\quad [\Phi]^{\mathrm{T}}[K][\Phi]=[K_r],\quad [K_r][M_r]^{-1}=[\Omega_r^2]$$

$[M_r]=\mathrm{diag}[m_1\ m_2\cdots m_n]$　　广义质量阵，模态质量矩阵

$[K_r]=\mathrm{diag}[k_1\ k_2\cdots k_n]$　　广义刚度阵，模态刚度矩阵

$\Omega_r^2=\mathrm{diag}[\Omega_1^2\ \Omega_2^2\cdots\Omega_n^2]$　　特征值矩阵

$[\Phi]=[\{\varphi_1\}\{\varphi_2\}\cdots\{\varphi_n\}]$　　固有振型矩阵，模态振型矩阵，模态矩阵

2.3 模态参数(Modal Parameters)

有阻尼固有频率、阻尼比、振型(Damped Natural Frequencies, Damping Ratios, Mode Shapes)

$$\omega_{dr} \quad \omega_r \qquad \zeta_r \qquad \{\phi\}_r$$

模态质量、刚度和阻尼(The Modal Mass, Stiffness and Damping)

$$m_r, \quad k_r, \quad c_r$$

阻尼系数和阻尼比(The Modal Damping Factor and Damping Ratio)

$$\sigma_r = \frac{c_r}{2m_r} \qquad \zeta_r = \frac{\sigma_r}{\omega_r}$$

2.4 阻尼矩阵的可对角化条件

Rayleigh阻尼，也称比例阻尼（Proportional Damping）是最常用的可对角化条件

$$C = \alpha M + \beta K$$

更一般的阻尼可对角化条件是Caughey于1965年提出的

$$C = M\sum_{j=0}^{N-1}\alpha_j[M^{-1}K]^j = M\sum_{j=0}^{N-1}\alpha_j A^j$$

更一般的阻尼可对角化条件也是Caughey(考伊)于1965年提出的

$$CM^{-1}K = KM^{-1}C$$

或写成

$$(M^{-1}C)(M^{-1}K) = (M^{-1}K)(M^{-1}C)$$

2.5 系统响应的模态叠加法(Modal Superimposition)

$$M\ddot{x}+C\dot{x}+Kx=f(t) \Leftarrow \begin{cases} x=Xe^{j\omega t}=\Phi q=\sum_{r=1}^{N} q_r\phi_r \\ q=Qe^{j\omega t},\quad f=Fe^{j\omega t} \end{cases}$$

模态坐标

$$Q=[diag(-\omega^2 m_r+\mathrm{j}\omega c_r+k_r)]^{-1}\Phi^{\mathrm{T}}F$$

系统响应的模态表示法

$$X=\Phi Q=\sum_{r=1}^{N} Q_r\phi_r$$

$$=\Phi[diag(-\omega^2 m_r+\mathrm{j}\omega c_r+k_r)]^{-1}\Phi^{\mathrm{T}}F$$

$$=\sum_{r=1}^{N}\frac{\phi_r\phi_r{}^{\mathrm{T}}F}{-\omega^2 m_r+\mathrm{j}\omega c_r+k_r}=H(\omega)F$$

$$x=Xe^{j\omega t}=\sum_{r=1}^{N} q_r\phi_r=\sum_{r=1}^{N} Q_r\phi_r e^{j\omega t}$$

第r阶模态贡献因子

$$Q_r=\frac{\phi_r{}^{\mathrm{T}}F}{-\omega^2 m_r+\mathrm{j}\omega c_r+k_r}$$

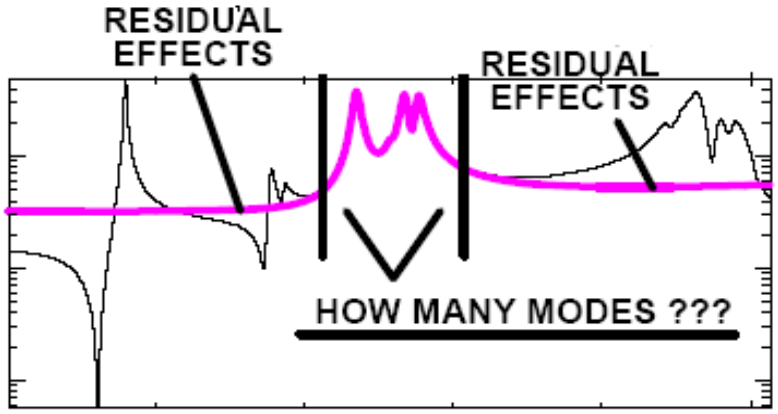
第 j点激励第 i点响应

$$X_i(\omega)=\sum_{r=1}^{N}\frac{\phi_{ir}\phi_{jr}}{-\omega^2 m_r+\mathrm{j}\omega c_r+k_r}F_j$$

模态模型

$$\{x\} = \sum_{r=1}^{N} q_r \{\varphi_r\}$$

- 运用模态参数建立响应计算模型，称为模态模型。
- 为减少计算量，实际计算中往往只考虑贡献大的模态（与激励方式有关）。



$$\{x\} = \sum_{r=m}^{n} q_r \{\varphi_r\}, \quad n - m < N$$

- 通过实验求得特征值和特征向量以及对应的其它模态参数，这就是实验模态分析的任务
- 模态分析法可以不必求所有的特征值和特征向量，只是求出所需要的那几阶，包括模态质量、模态刚度和模态阻尼，用以建立响应计算模型。

2.6 频响函数(FRFs)及其性质

频响函数矩阵H 定义

$$X = H(\omega)F$$

$$H(\omega) = \Phi(-\omega^2[m_r] + \mathrm{j}\omega[c_r] + [k_r])^{-1}\Phi^{\mathrm{T}}$$

分量形式

$$H_{ij}(\omega) = \frac{X_i(\omega)}{F_j(\omega)} = \sum_{r=1}^{N} \frac{\phi_{ir}\phi_{jr}}{-\omega^2 m_r + \mathrm{j}\omega\ c_r + k_r} = \sum_{r=1}^{N} \frac{\phi_{ir}\phi_{jr}}{m_r(\omega_r^2 - \omega^2 + 2\mathrm{j}\zeta_r\omega_r\omega)}$$

$$H_{ij}(\omega) = \frac{X_i(\omega)}{F_j(\omega)} = \sum_{r=1}^{N} \frac{\frac{1}{k_r}\phi_{ir}\phi_{jr}}{-\frac{\omega^2}{\omega_r^2} + 2\mathrm{j}\zeta_r\frac{\omega}{\omega_r} + 1}$$

H_{ij}第r阶模态的贡献

定义为j点单位激励力在i点引起的响应，实测H_{ij}时必须是单点激励引起的响应与该激励力之间在频域的比值。

H_{ij}第r阶模态的贡献

$$\left|{}_rH_{ij}(\omega)\right| = \frac{\frac{1}{k_r}\left|\phi_{ir}\phi_{jr}\right|}{\sqrt{\left[1-\frac{\omega^2}{\omega_r{}^2}\right]^2+(2\zeta_r\frac{\omega}{\omega_r})^2}}$$

∞ 共振 Max

$\zeta_r = 0, \quad \omega = \omega_r$

$\zeta_r \neq 0, \quad \omega = \omega_{dr}$

$\omega_{dr} = \omega_r\sqrt{1-2\zeta_r^2}$

有阻尼强迫振动共振频率

$$\zeta_r = \frac{\sigma_r}{\omega_r}$$

系统极点可由拉普拉斯域求解：极点方程

$$s^2m_r + sc_r + k_r = 0$$

$$s = a + jb$$

极点

$$\begin{cases} p_r = -\sigma_r + j\omega_{dr} \\ p^*{}_r = -\sigma_r - j\omega_{dr} \end{cases}$$

有阻尼自由振动频率

$$\omega_{dr} = \omega_r\sqrt{1-\zeta_r^2}$$

FRFs矩阵的性质

$$H(\omega)=\Phi diag[Y_r]\Phi^{\mathrm{T}}$$

第r阶导纳

$$Y_r(\omega)=\frac{1}{-\omega^2 m_r+\mathrm{j}\omega c_r+k_r}$$

对称性。在i点单位激励引起j点的响应与在j点单位激励引起i点的响应大小相等。又称Betti互易原理

$$H^{\mathrm{T}}=H,\quad H_{ij}=H_{ji}$$

FRFs矩阵包含系统全部模态信息。Y_r中包含m_r，k_r，c_r。Φ中包含全部模态振型

$$f_r=\frac{1}{2\pi}\omega_r=\frac{1}{2\pi}\sqrt{k_r/m_r}$$

$$\zeta_r=\frac{\sigma_r}{\omega_r}=\frac{c_r}{2m_r\omega_r}$$

FRFs矩阵的一行或一列同样包含系统的全部模态信息

$$\underset{j}{\mathrm{Col}}(H_{ij})=\sum_{r=1}^{N}Y_r\phi_{jr}\phi_r$$

$$\underset{i}{\mathrm{Row}}(H_{ij})=\sum_{r=1}^{N}Y_r\phi_{ir}[\phi_{1r}\quad\phi_{2r}\quad\cdots\quad\phi_{Nr}]=\sum_{r=1}^{N}Y_r\phi_{ir}\phi_r^{\mathrm{T}}$$

2.7 三种导纳曲线的比较

——位移导纳(Dynamic Compliance, Receptance)

幅频

$$Y_D = \frac{X}{F} = \frac{1/k}{\sqrt{[1-(\frac{\omega}{\Omega_n})^2]^2 + (2\zeta\frac{\omega}{\Omega_n})^2}}$$

相频

$$e^{j\Phi} = R_D + jI_D \qquad \Phi = -tg^{-1}\frac{2\zeta\frac{\omega}{\Omega_n}}{1-(\frac{\omega}{\Omega_n})^2}$$

实频

$$R_D = \frac{1-(\frac{\omega}{\Omega_n})^2}{k\{[1-(\frac{\omega}{\Omega_n})^2]^2 + (2\zeta\frac{\omega}{\Omega_n})^2\}}$$

虚频

$$I_D = \frac{-2\zeta\frac{\omega}{\Omega_n}}{k\{[1-(\frac{\omega}{\Omega_n})^2]^2 + (2\zeta\frac{\omega}{\Omega_n})^2\}}$$

位移导纳圆

$$R_D^2 + (I_D + \frac{1}{4\zeta k\frac{\omega}{\Omega_n}})^2 = (\frac{1}{4\zeta k\frac{\omega}{\Omega_n}})^2$$

$\omega = \Omega_n$ 时，圆心坐标 $(0, -\frac{1}{4\zeta k})$

—速度导纳(Mobility, Mechanical Admittance)

$$Y_V = \frac{V}{F} = \frac{\mathrm{j}\omega X}{F}$$

幅频

$$= \frac{1}{\omega m\sqrt{(2\zeta\frac{\Omega_n}{\omega})^2 + [1-(\frac{\Omega_n}{\omega})^2]^2}} e^{\mathrm{j}\alpha} = R_V + \mathrm{j}I_V$$

相频

$$\alpha = -\mathrm{tg}^{-1}\frac{1-(\frac{\Omega_n}{\omega})^2}{2\zeta(\frac{\Omega_n}{\omega})}$$

实频

$$R_V = \frac{2\zeta(\frac{\Omega_n}{\omega})}{\omega m\{(2\zeta\frac{\Omega_n}{\omega})^2 + [1-(\frac{\Omega_n}{\omega})^2]^2\}}$$

虚频

$$I_V = \frac{-[1-(\frac{\Omega_n}{\omega})^2]}{\omega m\{(2\zeta\frac{\Omega_n}{\omega})^2 + [1-(\frac{\Omega_n}{\omega})^2]^2\}}$$

速度导纳圆

$$(R_V - \frac{1}{2c})^2 + I_V{}^2 = (\frac{1}{2c})^2$$

or

$$(R_V - \frac{\Omega_n}{4k\zeta})^2 + I_V{}^2 = (\frac{\Omega_n}{4k\zeta})^2$$

圆心坐标 $(\frac{1}{2c}, 0)$

——加速度导纳(Accelerance)

$$Y_A = \frac{A}{F} = \frac{(\mathrm{j}\omega)^2 X}{F}$$

幅频

$$= \frac{1}{m\sqrt{[1-(\frac{\Omega_n}{\omega})^2]^2 + (2\zeta\frac{\Omega_n}{\omega})^2}} \mathrm{e}^{\mathrm{j}\beta} = R_A + \mathrm{j}I_A$$

相频

$$\beta = \mathrm{tg}^{-1}\frac{2\zeta\frac{\Omega_n}{\omega}}{1-(\frac{\Omega_n}{\omega})^2}$$

实频

$$R_A = \frac{1-(\frac{\Omega_n}{\omega})^2}{m\{[1-(\frac{\Omega_n}{\omega})^2]^2 + (2\zeta\frac{\Omega_n}{\omega})^2\}}$$

虚频

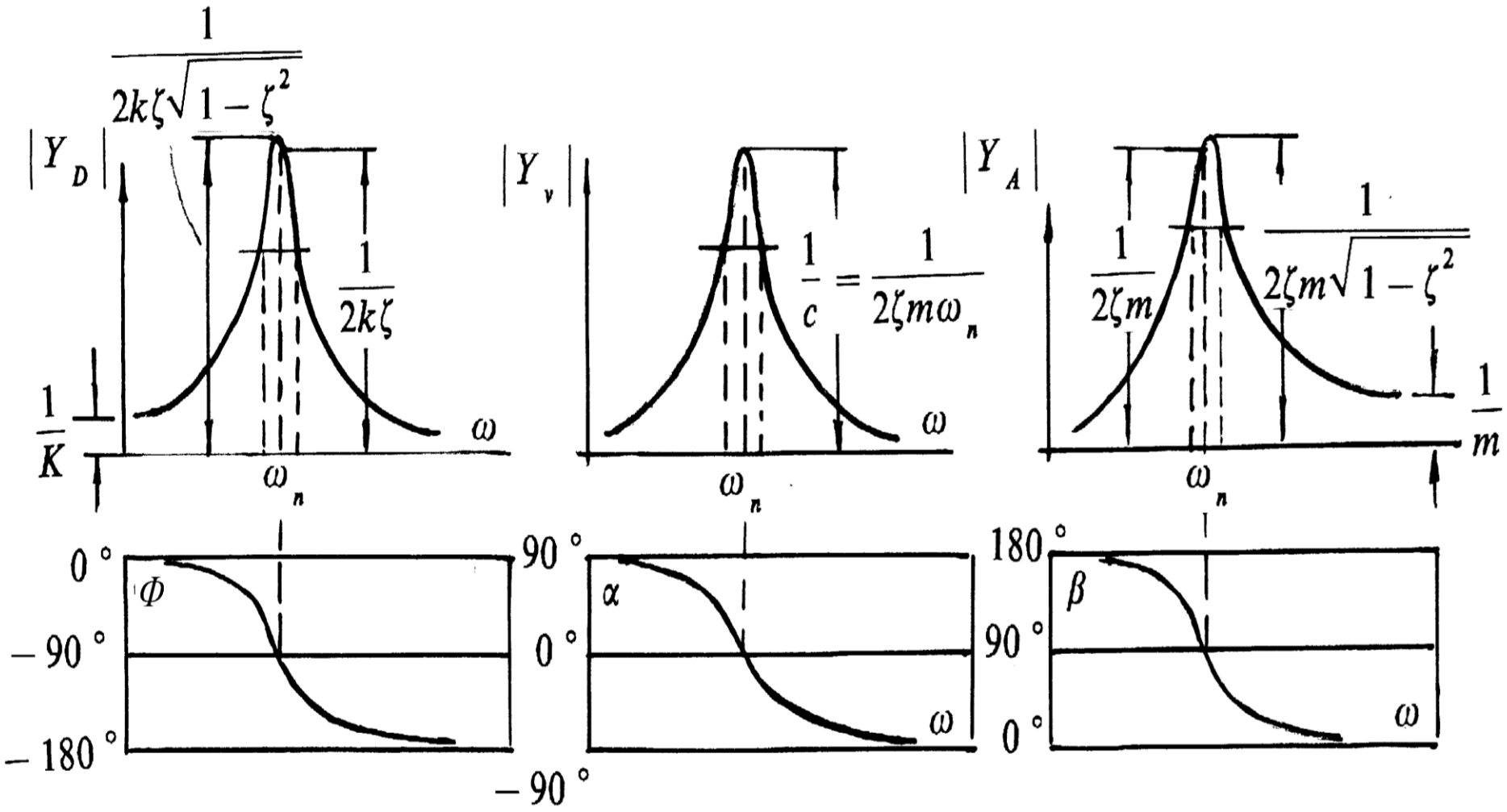
$$I_A = \frac{2\zeta\frac{\Omega_n}{\omega}}{m\{[1-(\frac{\Omega_n}{\omega})^2]^2 + (2\zeta\frac{\Omega_n}{\omega})^2\}}$$

加速度导纳圆

$$R_A^2 + (I_A - \frac{1}{4\zeta m\frac{\Omega_n}{\omega}})^2 = (\frac{1}{4\zeta m\frac{\Omega_n}{\omega}})^2$$

$\omega = \Omega_n$ 时，圆心坐标$(0, \frac{1}{4\zeta m})$

三种导纳的幅频及相频图



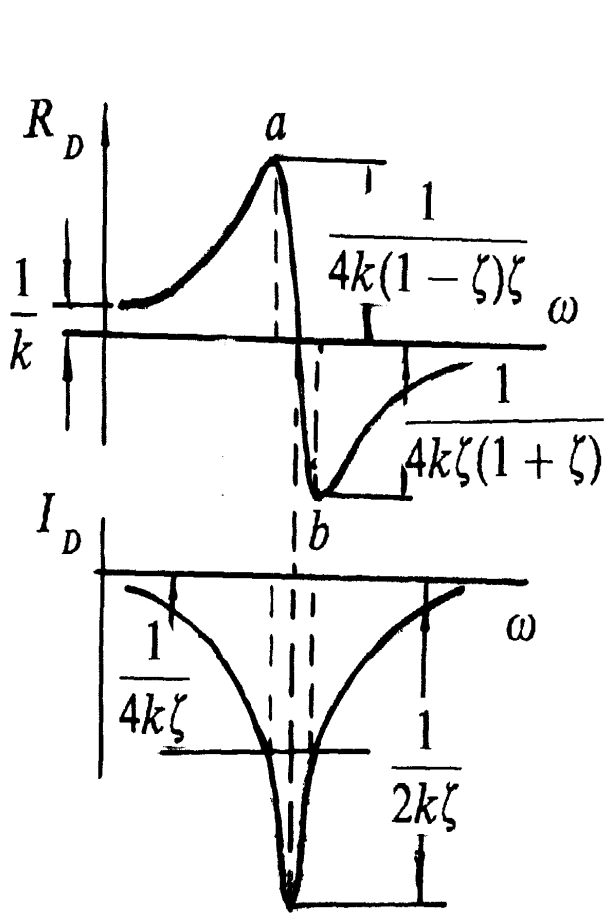
$$\omega_{dr}=\omega_r\sqrt{1-2\zeta_r^2}$$

$$\omega_{dr}=\omega_r$$

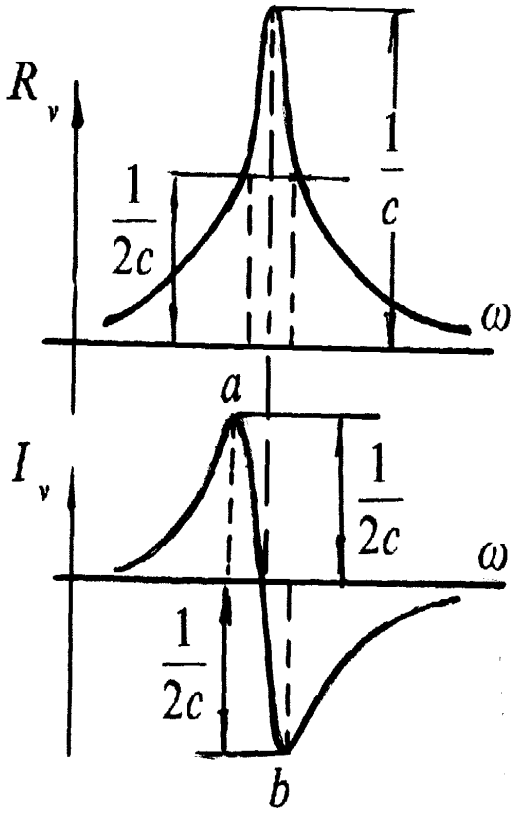
$$\omega_{dr}=\frac{\omega_r}{\sqrt{1-2\zeta_r^2}}$$

P116, P223

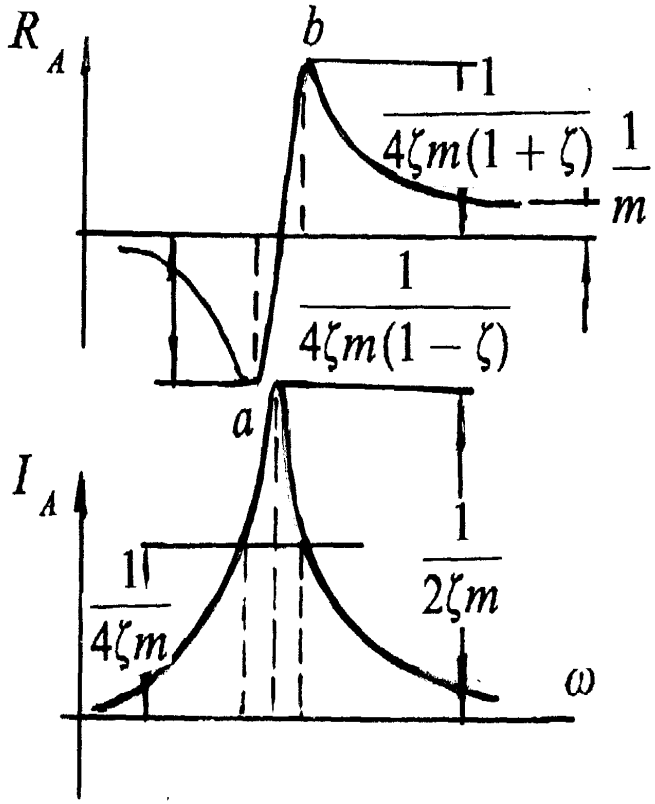
三种导纳的实频图及虚频图



$$\omega_{dr} \approx \omega_r \sqrt{1-\zeta_r^2}$$



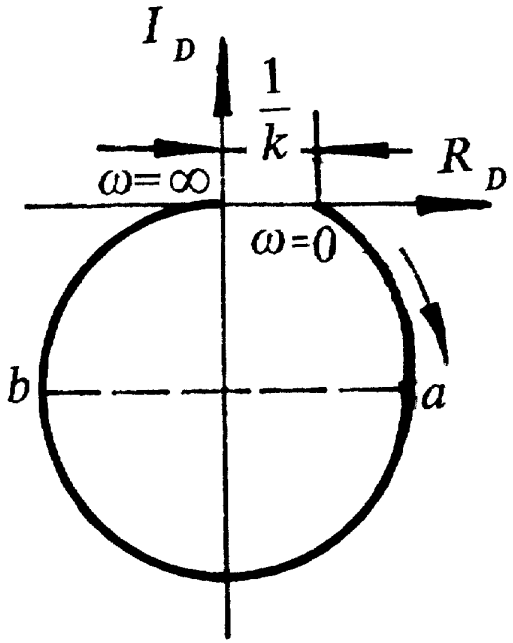
$$\omega_{dr} = \omega_r$$

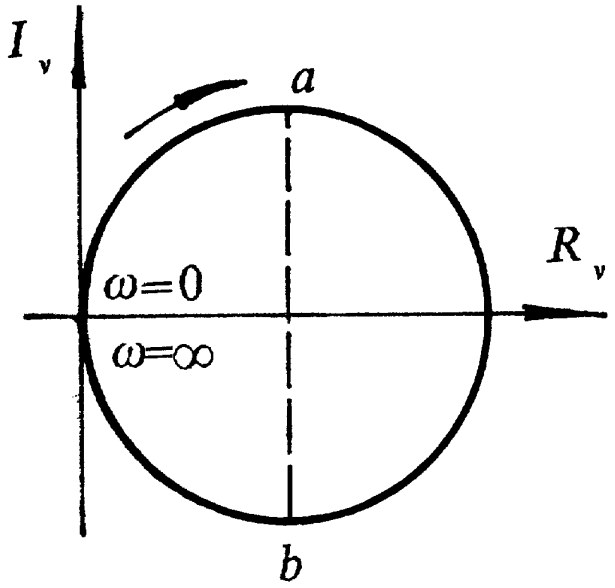


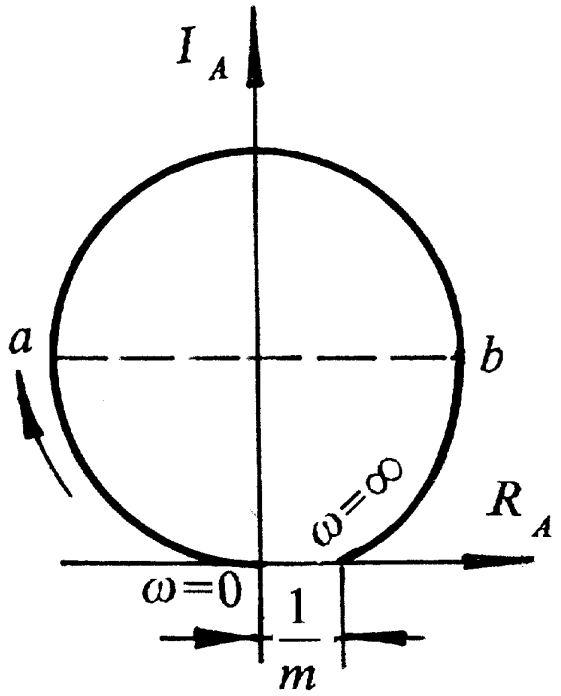
$$\omega_{dr} = \frac{\omega_r}{\sqrt{1-\zeta_r^2}} ?$$

P226

三种导纳圆(Nyquist Circle)







2.9 频响函数曲线分析

$$H_{ij}=\sum_{r=1}^{N}\frac{1}{k_r-\omega^2 m_r+j\omega c_r}\varphi_{ir}\varphi_{jr}=\sum_{r=1}^{N}Y_r\varphi_{ir}\varphi_{jr}=\sum_{r=1}^{N}{}_rH_{ij}$$

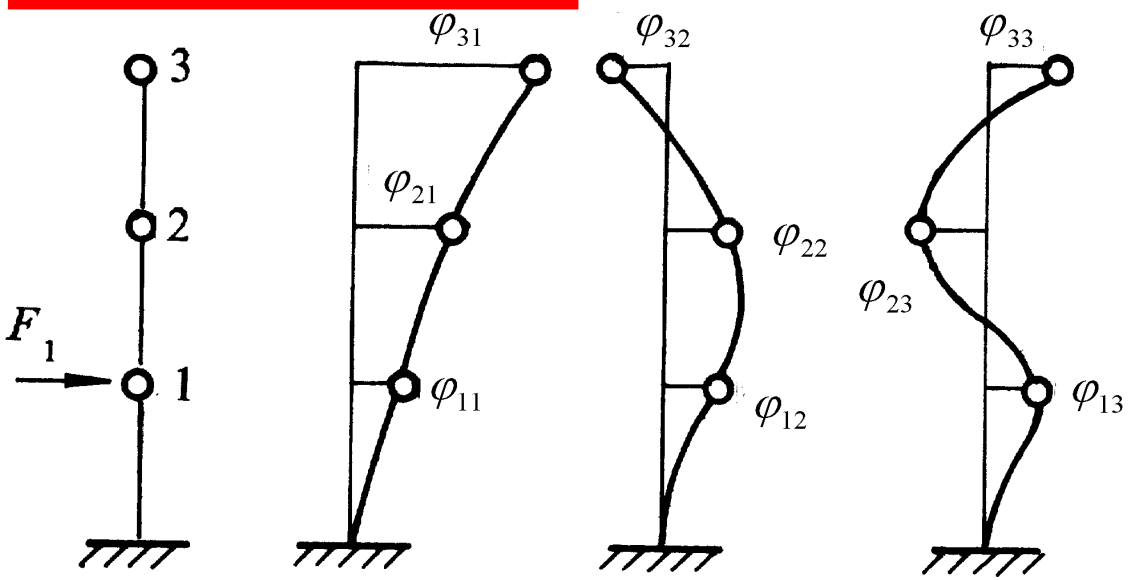
曲线形状主导项

$${}_rH_{ij}=Y_r\varphi_{ir}\varphi_{jr}=\frac{1}{k_r-\omega^2 m_r+j\omega c_r}\varphi_{ir}\varphi_{jr}$$

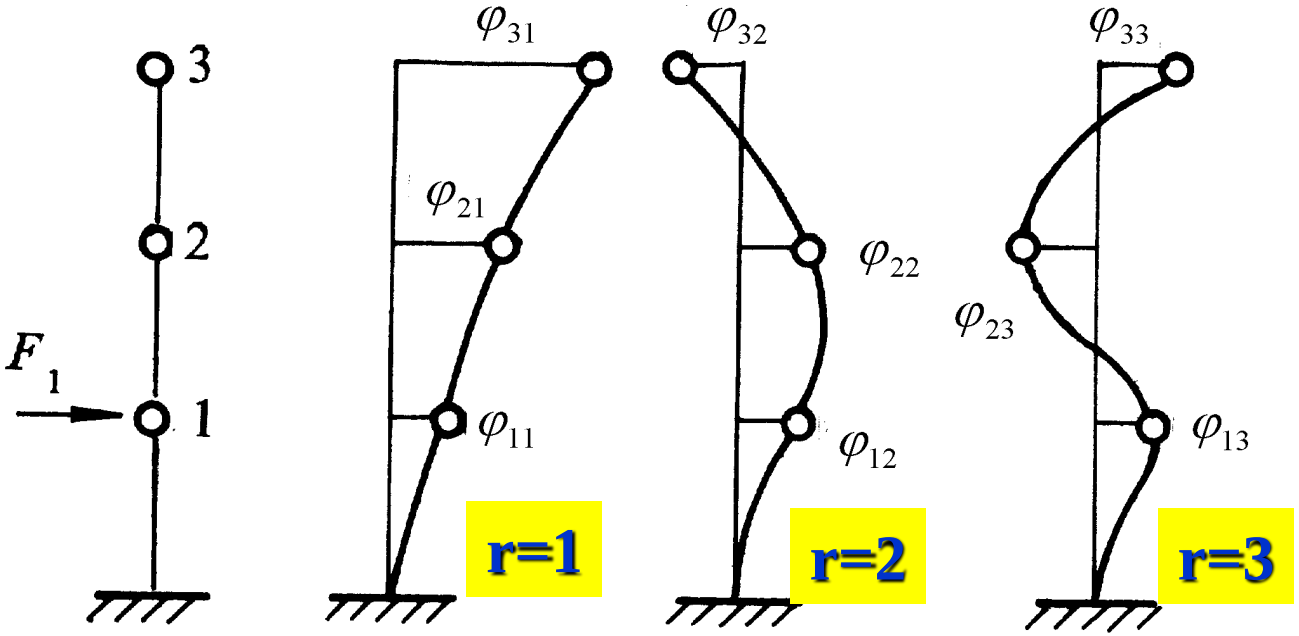
Example:

3 DOFs Model

频率稀疏分布时



$$H_{ij}=\sum_{r=1}^{N}Y_r\varphi_{ir}\varphi_{jr}$$



$$H_{11}={}_1H_{11}+{}_2H_{11}+{}_3H_{11}=Y_1\varphi_{11}^2+Y_2\varphi_{12}^2+Y_3\varphi_{13}^2=|H_{11}|e^{j\Phi_{11}}$$

$$=|Y_1|e^{j\Phi_1}\varphi_{11}^2+|Y_2|e^{j\Phi_2}\varphi_{12}^2+|Y_3|e^{j\Phi_3}\varphi_{13}^2$$

$$+\quad +\qquad +\quad +\qquad +\quad +$$

$$H_{21}={}_1H_{21}+{}_2H_{21}+{}_3H_{21}$$

$$=|Y_1|e^{j\Phi_1}\varphi_{21}\varphi_{11}+|Y_2|e^{j\Phi_2}\varphi_{22}\varphi_{12}+|Y_3|e^{j\Phi_3}\varphi_{23}\varphi_{13}=|H_{21}|e^{j\Phi_{21}}$$

$$+\quad +\qquad +\quad +\qquad -\quad +$$

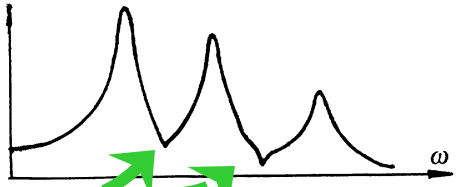
$$H_{31}={}_1H_{31}+{}_2H_{31}+{}_3H_{31}=|H_{31}|e^{j\Phi_{31}}$$

$$=|Y_1|e^{j\Phi_1}\varphi_{31}\varphi_{11}+|Y_2|e^{j\Phi_2}\varphi_{32}\varphi_{12}+|Y_3|e^{j\Phi_3}\varphi_{33}\varphi_{13}$$

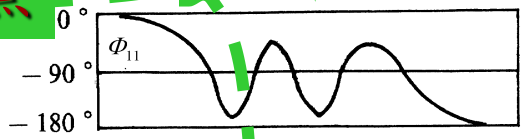
$$+\quad +\qquad -\quad +\qquad +\quad +$$

$$H_{11} = Y_1\varphi_{11}^2 + Y_2\varphi_{12}^2 + Y_3\varphi_{13}^2$$

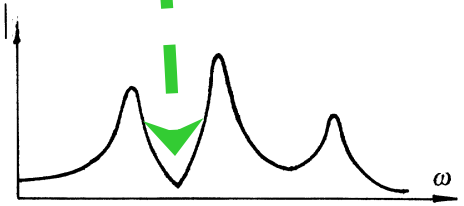
|H11|

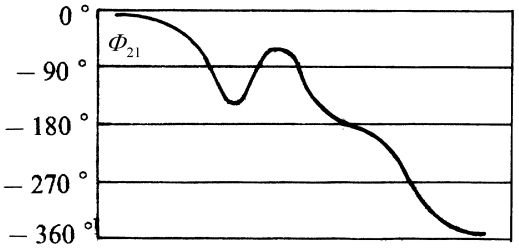


反共振点

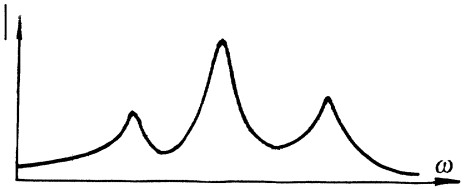


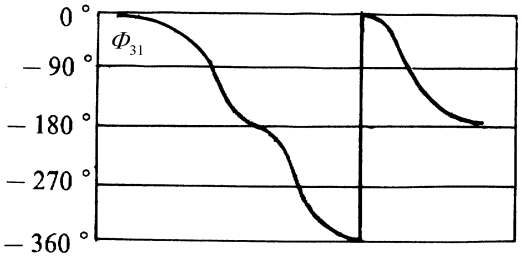
|H21|



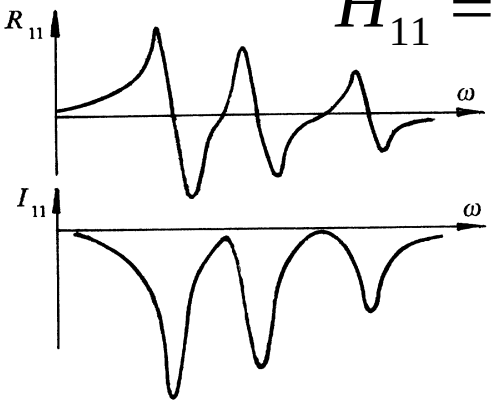


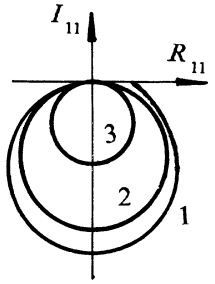
|H31|

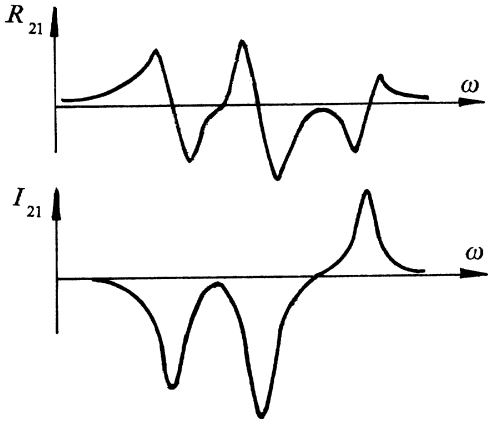


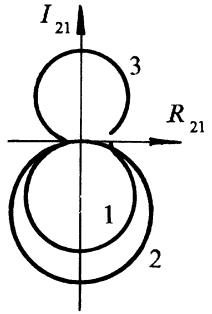


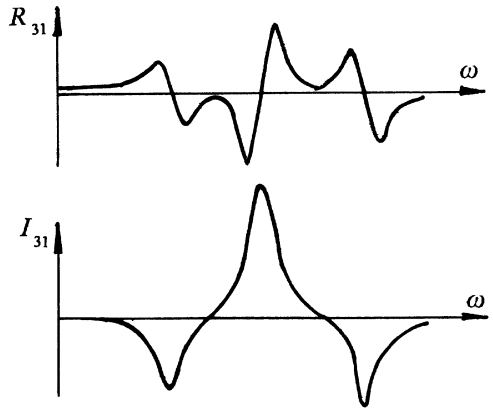
作业：
自行画出3DOF系统的频响函数定性曲线，并指出其中的特征点

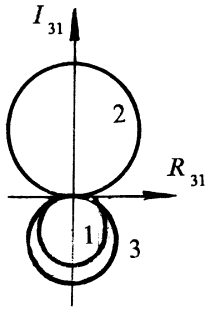












2.10 留数与振型

$$H_{ij}(\omega)=\sum_{r=1}^{N}\frac{\phi_{ir}\phi_{jr}}{m_r(\omega_r^2-\omega^2+2\mathrm{j}\zeta_r\omega_r\omega)}$$

$$=\sum_{r=1}^{N}\frac{R_{rij}\omega_{dr}}{\omega_r^2-\omega^2+2\mathrm{j}\zeta_r\omega_r\omega}$$

即：

$$\frac{\phi_{ir}\phi_{jr}}{m_r}=\omega_{dr}R_{rij}$$

第r阶留数矩阵

$$R_r=\frac{1}{m_r\omega_{dr}}\{\phi\}_r\{\phi\}_r^{\mathrm{T}}$$

$$m_r=\frac{1}{a_r\omega_{dr}}$$

$$a_r=\frac{1}{m_r\omega_{dr}}=\frac{R_{rij}}{\phi_{ir}\phi_{jr}}$$

$$k_r=\frac{\sigma_r^2+\omega^2{}_{dr}}{a_r\omega_{dr}}$$

$$c_r=\frac{2\sigma_r}{a_r\omega_{dr}}$$

3 模态振型规一化方法(Normalization of Mode Shapes)

由频响函数数据可以得到

$$R_r = \frac{1}{m_r \omega_{dr}} \{\phi\}_r \{\phi\}_r^{\mathrm{T}}$$

留数 R_{rij}

频率 $\omega_{dr} = \omega_r \sqrt{1-\zeta_r^2}$

阻尼 $\sigma_r = \frac{c_r}{2m_r}$ $\zeta_r = \frac{\sigma_r}{\omega_r}$ $c_r = \frac{2\sigma_r}{a_r \omega_{dr}}$

但无法得到全部的模态参数。通过模态振型规一化可以得到全部模态参数，常见模态振型规一化方法包括：

$m_r = \frac{1}{a_r \omega_{dr}} = 1$ $a_r = 1$

$\phi_{jr} = 1$ 模态振型的模取1

$\phi_{pr} = 1$,p为激励点

模态质量规一

$$m_r = \frac{1}{a_r \omega_{dr}} = 1$$

$$R_{rij} = a_r \phi_{ir} \phi_{jr} = \frac{\phi_{ir} \phi_{jr}}{m_r \omega_{dr}}$$

$$a_r = \frac{1}{m_r \omega_{dr}} = \frac{1}{\omega_{dr}}$$

激励点**j**

$$R_{rjj} = a_r \phi_{jr} \phi_{jr}$$

$$\phi_{jr} = \sqrt{\frac{R_{rjj}}{a_r}} = \sqrt{R_{rjj} \omega_{dr}}$$

$$R_{rij} = a_r \phi_{ir} \phi_{jr}$$

$$R_{rij} = a_r \phi_{ir} \sqrt{\frac{R_{rjj}}{a_r}} = \sqrt{\frac{R_{rjj}}{\omega_{dr}}} \phi_{ir}$$

响应点**i**

$$\phi_{ir} = \sqrt{\frac{\omega_{dr}}{R_{rjj}}} R_{rij}$$

激励点j振型元素为1

$$\phi_{jr}=1$$

$$R_{rij}=a_r\phi_{ir}\phi_{jr}=\frac{\phi_{ir}\phi_{jr}}{m_r\omega_{dr}}$$

激励点j

$$R_{rjj}=a_r\phi_{jr}\phi_{jr}=a_r$$

$$R_{rij}=a_r\phi_{ir}\phi_{jr}=a_r\phi_{ir}$$

响应点i

$$\phi_{ir}=\frac{R_{rij}}{a_r}=\frac{R_{rij}}{R_{rjj}}$$

任意点p振型元素为1

$$\phi_{pr}=1$$

$$R_{rij}=a_r\phi_{ir}\phi_{jr}=\frac{\phi_{ir}\phi_{jr}}{m_r\omega_{dr}}$$

激励点j

$$R_{rjj}=a_r\phi_{jr}\phi_{jr} \Rightarrow \phi_{jr}=\sqrt{\frac{R_{rjj}}{a_r}}$$

$$R_{rpj}=a_r\phi_{pr}\phi_{jr}=a_r\phi_{jr}=a_r\sqrt{\frac{R_{rjj}}{a_r}}=\sqrt{a_rR_{rjj}} \Rightarrow a_r=\frac{R_{rpj}^2}{R_{rjj}}$$

$$R_{rij}=a_r\phi_{ir}\phi_{jr} \Rightarrow \frac{R_{rij}}{R_{rpj}}=\frac{a_r\phi_{ir}\phi_{jr}}{a_r\phi_{pr}\phi_{jr}}=\frac{\phi_{ir}}{\phi_{pr}}=\phi_{ir}$$

$$\phi_{ir}=\frac{R_{rij}}{R_{rpj}}$$

响应点i

比例系数为1

$$a_r = 1$$

$$R_{rij} = a_r \phi_{ir} \phi_{jr} = \frac{\phi_{ir} \phi_{jr}}{m_r \omega_{dr}}$$

激励点**j**

$$\phi_{jr} = \sqrt{\frac{R_{rjj}}{a_r}} = \sqrt{R_{rjj}}$$

$$R_{rij} = a_r \phi_{ir} \phi_{jr} = \sqrt{R_{rjj}}\, \phi_{ir}$$

响应点**i**

$$\phi_{ir} = \frac{1}{\sqrt{R_{rjj}}} R_{rij}$$

模态振型的模取1

$$R_{rij} = a_r \phi_{ir} \phi_{jr} = \frac{\phi_{ir}\phi_{jr}}{m_r \omega_{dr}}$$

$$\sqrt{\sum_{i=1}^{N} \phi_{ir}^{\ 2}} = 1$$

$$1 = \sqrt{\sum_{i=1}^{N} \phi_{ir}^{\ 2}} = \sqrt{\sum_{i=1}^{N} \left(\frac{R_{rij}}{\sqrt{a_r R_{rjj}}} \right)^2} = \sqrt{\sum_{i=1}^{N} \frac{R_{rij}^2}{a_r R_{rjj}}} = \sqrt{\frac{1}{a_r R_{rjj}} \sum_{i=1}^{N} R_{rij}^2}$$

激励点j

$$\phi_{jr} = \sqrt{\frac{R_{rjj}}{a_r}}$$

$$\phi_{ir} = \frac{R_{rij}}{a_r \phi_{jr}} = \frac{R_{rij}}{\sqrt{a_r R_{rjj}}}$$

$$a_r = \frac{1}{R_{rjj}} \sum_{i=1}^{N} R_{rij}^2$$

响应点i

$$\phi_{ir} = \frac{R_{rij}}{\sqrt{a_r R_{rjj}}} = \frac{R_{rij}}{\sqrt{\frac{R_{rjj}}{R_{rjj}} \sum_{i=1}^{N} R_{rij}^2}} = \frac{1}{\sqrt{\sum_{i=1}^{N} R_{rij}^2}} R_{rij}$$

4 基于速度或加速度信号的传递函数及模态参数

加速度、速度和位移留数之间的关系：

$$R^{(a)} = 2\pi f_r R^{(v)} = (2\pi f_r)^2 R^{(d)}$$

(a)、(v)、(d)分别指加速度、速度和位移

Literature for further reading

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- 郑兆昌、潭明一，多自由度系统复模态理论的摄动法（二）—重特征值及高阶摄动，应用力学学报，1985,2(4):17-31